A Cloud Detection Algorithm for Early Morning Observations From the FY-3E Satellite

Ni An, Huazhe Shang, Wei Lesi, Xu Ri, Chong Shi, Gegen Tana, Yuhai Bao, Zhaojun Zheng, Na Xu, Lin Chen, Peng Zhang, *Fellow, IEEE*, Lingmeng Ye, and Husi Letu, *Senior Member, IEEE*

***Abstract*— Accurate cloud detection via satellites is important for cloud radiative forcing estimation and disaster weather monitoring. Current polar-orbiting satellite cloud observations are limited during early morning orbit and contain notable uncertainty due to dimness measurements in visible bands. FY-3E\MERSI-LL is the first early morning orbit satellite worldwide and can realize global cloud observation under early morning scenarios. In this study, a dynamic threshold cloud detection algorithm is proposed based on the FY-3E\MERSI-LL infrared channel, combined with auxiliary data such as sea surface temperature (SST), land surface temperature (LST), snow cover mask (SCM), and terrain elevation. The algorithm can detect clouds against a complex land surface background, but faces classification difficulties over some plateau, high-latitude, and snow surface regions, especially during early morning observation periods. Compared to coincident Himawari-8 and Geostationary Operational Environmental Satellite (GOES)-16 cloud measurements in the Eastern and Western Hemispheres, respectively, our algorithm recognizes reasonable cloud distributions. Furthermore, Himawari-8 and GOES-16 cloud products are used for quantitative cloud algorithm evaluation. The results show that at low-middle latitudes (60°N–60°S), the average cloud and clear hit rates (HRs) during the various seasons are 73.24% and 76.46%, respectively, the cloud leakage and false alarm rates (FARs) are 14.46% and 8.15%, respectively, and the total accuracy (cloud and clear) is 77.33%. The algorithm performance is better over the ocean than over land. Ground site MPLCMASK products are also used to verify the FY-3E cloud results in middle- and high-latitude areas. This algorithm provides a cloud detection reference during early morning orbit based on infrared channels.**

***Index Terms*— Cloud detection, cloud detection accuracy, cloud detection algorithm, dynamic threshold, early morning orbit, FY-3E\MERSI-LL.**

I. Introduction

RELIABLE cloud distributions are directly related to cloud radiation forcing and climate research and monitoring [1], [2], [3]. As early as 1983, the International Satellite Cloud Climate Project (ISCCP) was established to observe the impact of clouds on the water cycle to further understand the global radiation energy budget and to parameterize clouds in climate models [4], [5], [6]. At the same time, the latest climate change report of the 6th Intergovernmental Panel on Climate Change (IPCC6) confirmed that the radiation characteristics of clouds are the largest source of uncertainty in the overall climate feedback [7],.() [8]. Therefore, accurate cloud distribution data can better explain climate models.

Satellite remote sensing technology is one of the most effective methods for observing global clouds to predict weather and monitor climate change [9]. Cloud detection algorithms can be classified based on surface type to distinguish between day and night, but polar-orbiting satellites cannot provide early morning observations, such as Moderate Resolution Imaging Spectrometer (MODIS) Terra\Aqua global cloud products, in which cloud detection algorithm application scenarios are divided into ocean, land, ice, desert, etc., and into day and night for the entire globe [10], [11], [12], [13], the Clouds and Earth's radiant energy system CERES cloud detection has been developed considered a variety of clear conditions during daytime and nighttime [14], [15], [16]. While in the Clouds from the Advanced Very High Resolution Radiometer (AVHRR) Extended System (CLAVR-x) (latest version updated in 2013), cloud detection algorithms were developed considering the radiance or associated brightness temperature based on AVHRR data during the day and night [17], [18], [19], [20]. In the meantime, the Cloud and Aerosol Imager (CAI) is carried by the Greenhouse Gases Observing Satellite (GOSAT), and its threshold tests and threshold values can be adjusted according to surface type, such as water, land, and polar region [21]. Moreover, certain geostationary satellites can realize continuous cloud detection including early morning periods, and are not constrained

Manuscript received 29 April 2023; revised 7 July 2023; accepted 6 August 2023. Date of publication 15 August 2023; date of current version 29 August 2023. This work was supported in part by the National Science Fund for Distinguished Young under Grant 42025504, in part by the Youth Innovation Promotion Association of the Chinese Academy of Sciences under Grant 2021122, in part by the National Natural Science Foundation of China under Grant 42175152, and in part by the Fengyun Application Pioneering Project: Development of FY-3E MERSI-LL Cloud Detection Algorithm under Grant FY-APP-2021.0505. *(Corresponding authors: Huazhe Shang; Husi Letu.)*

Ni An is with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100101, China, also with the College of Geographical Sciences, Inner Mongolia Normal University, Hohhot 010020, China, and also with the College of Resources and Environment, Baotou Teachers' College, Baotou 014030, China (e-mail: anni@bttc.edu.cn).

Huazhe Shang, Wei Lesi, Chong Shi, and Husi Letu are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100101, China (e-mail: shanghz@radi.ac.cn; weils@aircas.ac.cn; shichong@aircas.ac.cn; husiletuw@hotmail.com).

Xu Ri is with the College of Geography and Environmental Science, Northwest Normal University, Lanzhou 730070, China, and also with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100101, China (e-mail: xuri916924@163.com).

Gegen Tana is with the National Space Science Center, Chinese Academy of Sciences, Beijing 100190, China.

Yuhai Bao is with the College of Geographical Sciences, Inner Mongolia Normal University, Hohhot 010020, China.

Zhaojun Zheng, Na Xu, Lin Chen, and Peng Zhang are with the National Satellite Meteorological Center, Beijing 100081, China (e-mail: xuna@cma.gov.cn).

Lingmeng Ye is with Huayun ShineTek Technologies Company Ltd., Beijing 100081, China.

Digital Object Identifier 10.1109/TGRS.2023.3304985

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:44:57 UTC from IEEE Xplore. Restrictions apply.

by observation period, such as the Himawari-8\AHI cloud products in daytime in Japan-centered areas [22], [23], [24] and the Geostationary Operational Environmental Satellite (GOES)\Advanced Baseline Imager (ABI) cloud mask (AMC) products [25], [26], in most of the Western Hemisphere. These cloud products can realize real-time cloud observations in different hemispheric regions but are limited within the designated observation range. Early morning observation periods are necessary and can function as the day and night bridge to ensure cloud detection continuity and accuracy. In view of this demand, research has been conducted, Meteosat Second Generation (MSG) developed cloud detection algorithms via the addition of dusk time [27]. However, whether these cloud tests can be better applied to FY-3E during early morning orbit must be carefully considered from FY-3E six infrared channel perspective and should be further investigated.

Considering the polar-orbit meteorological satellite spatial and temporal integrity of the observation range and repetition period for clouds, a series of morning and afternoon orbit satellites have been built, such as the European MetOp satellites for morning/afternoon orbits and the American Joint Polar Satellite System (JPSS) satellite for afternoon orbit, but they lack an early morning orbit leaving a gap of 2–3 orbits per 6 h global observation assimilation time window. To fill this observation gap, some integrated products are needed for incorporation to achieve a large range of synchronous observations covering this period gap, such as Ozone Monitoring Instrument (OMI)\Aura and MODIS\Aqua cloud products, which can be almost simultaneously observation combined to realize the complementary advantages of the different satellite data, but they suffer from errors due to the time lag between the satellites and merged data [28]. In addition, with stationary satellite data, the EUMETSAT Meteorological Product Extraction Facility (MPEF) fuses Meteosat 0°, Meteosat Indian Ocean Data Coverage (IODC), GOES-16, GOES-17, and Himawari-8 data to develop RGB Fog\Low cloud products with a limited observation time [29]. To more directly solve these problems, the Chinese FY-3E satellite, the world's first early morning orbiting satellite launched in 2021, with FY-3A\C and FY-3B\D satellites in accordance with morning and afternoon orbits, respectively, achieved the three-satellite network proposed by the World Meteorological Organization (WMO). This observation network can accomplish a worldwide 100% coverage of numerical weather prediction (NWP), bridging the observation time-gap of polar orbit satellites, and shortening the repeat observation period. Moreover, the advantages of three-satellite network observations have been demonstrated in microwave soundings of MetOp-A, NOAA-18, and NOAA-15 satellites [30], while Joo et al. [31] also confirmed that three-satellite network observations could achieve continuous orbit coverage of overlapping observations at high latitudes and a 4-h coverage at low latitudes. Therefore, cloud observations based on the FY-3E satellite during early morning orbit combined with FY-3A\C and FY-3B\D data are greatly important worldwide.

The first early morning orbit satellite FY-3E\MERSI-LL provides one low-light channel and six infrared channels. However, due to the limited solar illumination during this period and the absence of visible channels, cloud detection becomes highly uncertain. Consequently, it is difficult and challenging to implement the cloud detection algorithm for complex surfaces worldwide in various seasons, using FY-3E data with missing visible channels and fewer detection channels. In this article, a cloud detection algorithm is developed that combines spectral signature threshold and dynamic threshold methods to classify data as cloud, probably cloud, clear, and probably clear categories, which is applied to complex surface backgrounds, such as terrains with different elevations, snow-covered surfaces, and different times of the year. Finally, comparisons to other cloud datasets are presented to demonstrate the performance of the algorithm. Therefore, this article is organized as follows, data and cloud detection methods are introduced in Sections II and III, an analysis and verification of the cloud detection results are provided in Section IV, and we summarize the conclusions and present a discussion in Section V.

II. Data

In this study, FY-3E\MERSI-LL Level-1 data are mainly used to execute the cloud detection algorithms with FY-3E auxiliary data land cover data (LC) data and digital elevation mode (DEM), as well as level-2 sea surface temperature (SST), land surface temperature data (LST), and snow cover mask (SCM) products. Furthermore, Himawari-8\AHI and GOES-16\ABI 10.4 μm channel brightness temperature images and MOD35 global cloud products are used for comparison and qualitative verification. Himawari-8, GOES-16 cloud products, and ground lidar/radar data Micropulse Lidar Cloud Mask (MPLCMASK) value-added products are also used for real-time quantitative accuracy verification.

A. FY-3E Level-1 Data

The world's first early morning orbit meteorological satellite FY-3E was successfully launched on July 5, 2021, with a local landing point (ETC) at approximately 6:00 P.M. The FY-3E satellite, comprising the FY-3A\C and FY-3B\D satellites provides a three-orbit network system worldwide, with observation capability during early morning, mid-morning, and afternoon orbits [32]. The Medium Resolution Spectral Imager-Low-Light (MERSI-LL) instrument carried by FY-3E provides seven channels, including one low-light channel and six thermal infrared channels, among which two infrared channels have a 250 m spatial resolution, and the others exhibit a 1000 m spatial resolution as shown in Table I. The brightness temperature of infrared channels is applied to distinguish between cloud and clear areas in detection tests. Additional FY-3E LC and DEM data are also employed in cloud detection algorithms over plateaus and desert surfaces [33].

B. Auxiliary Data

Because a variety of continental surface types have different brightness temperature values under clear sky conditions, LST and SST data can be used as auxiliary data for cloud detection algorithms. More information on the FY-3E Level-2 products, namely SST, LST, and SCM is given in Table II. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:44:57 UTC from IEEE Xplore. Restrictions apply.

TABLE I
FY-3E\MERSI-LL CHANNEL INFORMATION

Channel	Wavelength (μm)	Maximum radiation L_{max}/T_{max} W/m²/sr (K)	Minimum radiation L_{min}/T_{min} W/m²/sr (K)	Typical radiation L_{typ}/T_{typ} W/m²/sr (K)	Resolution (m)
CH1	0.70	90	3e-5	4e-5 (night) 50 (day)	1000
CH2	3.8	350	186	300	1000
CH3	4.05	380	185	300/380	1000
CH4	7.2	270	186	270	1000
CH5	8.55	330	185	270	1000
CH6	10.8	345	185	300	250
CH7	12.0	345	185	300	250

TABLE II
CLOUD DETECTION AUXILIARY DATA INFORMATION

Name	Format	Repeated observation time	Spatial resolution	Application scenes
SCM	HDF	24 h	0.05°	Snow and ice
SST	HDF	24 h	0.05°	Sea
LST	HDF	5 min	0.25°	Land
LC	HDF	5 min	0.01°	Desert
DEM	HDF	5 min	0.01°	Sea and land

LST products are simulated with the MODTRAN radiation transmission model retrieved by the split window inversion algorithm, the observation interval is divided into one day, ten days, and one month respectively, and the spatial resolutions are 0.0025° and 0.01°. The SST products are produced by algorithms using 11 and 12 μm channel brightness temperatures and fused into daily and monthly products with 5-min observation intervals and a 5-km spatial resolution. The LST and SST data are used for cloud detection in ocean and land areas, respectively. The SCM data used in this study are global products based on the fusion of FY satellite optical remote sensing snow cover and microwave remote sensing snow depth data. The products include 1, 2, and 3-day synthetic snow (including sea ice) cover data with a 0.05° spatial resolution, and snow data have been introduced into the MODIS cloud product algorithm [34], which is used to detect clouds over snow and ice land surfaces.

C. Validation Data

Considering the coincidence in time and location, MOD35 global cloud products are used for comparison; the Collection 6 MOD35 global cloud products are applied to the spectral cloud detection algorithm under various scenarios, which can better realize the cloud detection under different atmospheric conditions, but time error is large during different orbits, which is a drawback; geostationary satellite cloud data are also collected for verification, Himawari-8\AHI was designed with 16 spectral channels ranging from 0.47 to 13.3 μm with a 5 km spatial resolution, and the full-disk observation interval is 10 min (ftp.ptree.jaxa.jp). On the other side of the hemisphere, GOES-16 with an ABI sensor is equipped with 16 spectral channels from visible light to infrared light, and 5–15 min observation intervals with a 2–5 km spatial resolution. The observation area is divided into full-disk region, American region, and mesoscale region (ftp://goes3g-est-nc.cr.chiba-u.ac.jp). Therefore, the 10.4 μm channel brightness temperature continuously observed by these two satellites can be used to verify the cloud distribution in the Eastern and Western Hemispheres. The observation interval of the Himawari-8 CARE 24-h cloud product data observation interval is 10 min, and 24 h cloud detection is suitably realized through detection algorithms such as threshold methods and machine learning methods [35]. GOES-16\ACM provides the official binary clear-sky mask (clear or cloudy), and provides a four-level cloud mask (clear, probably clear, probably cloud, and cloud) with nine spectral band signature tests.

Lidar and radar are effective synergistic systems to provide more complete cloud products than any one system. The ground site data MPLCMASK value-added product can provide cloud information at 30 m and 30 s resolutions applied between 0.5 and 20 km above ground level, including the number of cloud layers, each layer's cloud top and cloud base as well as cloud boundaries, and using the methodology developed by Flynn et al. [36] and Campbell et al. [37]. Its active sites of the Eastern North Atlantic (ENA central location: 39°5′29.76″ N, 28°1′32.52″ W), North Slope Alaska (NSA, central location: 71°19′22.8″ N, 156°36′54″ W) and the Southern Great Plains (SGP, central location: 36°36′26.3592″ N, 97°29′15.5148″ W) are selected to verify the accuracy of cloud detection. The ENA atmospheric observatory is the newest measurement site, and the SGP atmospheric observatory was the first field measurement site established by the atmospheric radiation measurement (ARM) user facility. At the same time, The NSA atmospheric observatory can provide comprehensive data about cloud and radiative processes at high latitudes. The MPLCMASK value-added product sampling interval is 30 s, which perfectly matches the 5 min time resolution of the FY-3E Level-1 data, and the sites are spatially matched to the image data within 1000 m spatial resolution image pixels of the FY-3E data where sites are located.

ERA5 can provide hourly estimates for a large number of atmospheric, ocean-wave, and land-surface quantities. Its hourly data on pressure levels has been regridded to a regular lat-lon grid of 0.25°, and the vertical coverage range is from 1000 to 1 hPa divided into 37 pressure levels. The main variable hourly temperature data can be used to explore the characteristics of temperature differences between the land surface and cloud tops in polar and equatorial regions.

III. METHODS

Clouds are generally more reflective in the visible but colder than the background as measured in the thermal infrared channel, and clear thermal radiation properties are determined by the background surface. The different application scenarios over various surface types and the corresponding spectral responses in the sensor spectral channel result in diverse cloud detection algorithms. A threshold method is proposed to distinguish clouds from the clear sky by using the satellite's visible reflectance and infrared radiation affected by solar illumination, atmospheric conditions, and surface type combined

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:44:57 UTC from IEEE Xplore. Restrictions apply.

TABLE III
FY-3E CLOUD DETECTION ALGORITHMS BASED ON INFRARED CHANNEL IN THE DIFFERENT GROUPS

Group	Tests	Application scenes	Ancillary input	Weight
Group 1	I3.8, I7.2, I10.8, I12.0;	All		1
	ID7.2_10.8;	All		1
Group 2	(BT3.8, BT10.8) & 2500<DEM<3000;	Plateau	DEM	1
	BTD8.6_7.2 & DEM>3000;	All	DEM	1
	Dynamic threshold LUT: BT10.8;	All		1
	(LST-BT12.0, BT12.0, BTD72_12.0) & DEM>0;	Land	LST, DEM	2
	(SST-BT10.8, BT3.8, BT10.8, BT12.0) & DEM<=0;	Sea	SST, DEM	1
Group 3	(BT3.8, BT10.8, BT12.0) & 2500<DEM<3000;	Plateau	DEM	1
	(BT3.8, BT10.8, BT12.0) & 3000<DEM;	Plateau	DEM	1
	Dynamic threshold LUT: BT10.8 & 2500<DEM;	Plateau	DEM	1
	I10.8 & DEM≤0 & LC;	Desert	DEM, LC	1
Group 4	BT10.8;	All		1
	BTD10.8_12.0;	All		2
	BTD7.2_10.8 & SCM;	Snow	SCM	2
	BTD10.8_3.8;	All		2
	BTD8.6_7.2.	All		1

Notes: I** is the normalized channel brightness temperature (K), and ID** is the normalized brightness temperature difference. ** is the corresponding wavelength (μm), BT** is the brightness temperature of the * * channel, BTD**_** is the brightness temperature difference of the two bands (K), and & is the union of the test results.

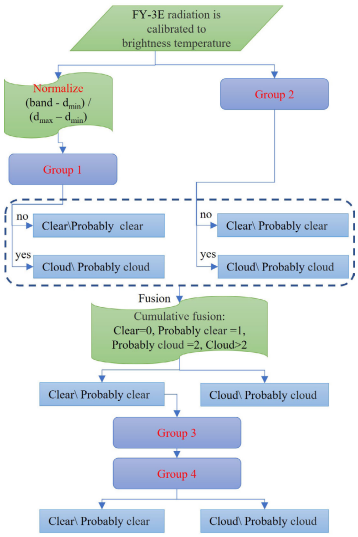


Fig. 1. FY-3E cloud detection algorithm flowchart, where BT indicates the channel brightness temperature, and BTD indicates the difference value in the channel brightness temperature.

with the LST, SST, and SCM data and auxiliary LC and DEM data. The FY-3E polar-orbiting satellite, with fewer detection channels, supports a cloud detection method combining the threshold and dynamic threshold methods to achieve four types of cloud detection results in different scenes worldwide: cloud, probably cloud, clear, and probably clear.

Because some threshold tests have been applied in the study of certain clouds, and threshold values are associated with specific scenes. However, the difference in radiance between cloud and clear in the low-light channel is not very significant, and the low-light signal is less stable and sometimes is weak, so the detection tests mainly use the thermal infrared channels and are divided into four Groups with different conditions as shown in Fig. 1, and the detailed tests are listed in Table III [38]. The impact of these test results on the final outcome is taken into consideration, some tests are assigned a weight of 2, which have a higher detection accuracy and higher precision or are less affected by latitude changes and seasonal changes, and other tests are assigned a weight value of 1.

In the first Group, to enhance the brightness temperature difference between clouds and clear at high latitudes and to avoid the boundary connection problem between images, the channel brightness temperatures are normalized, and the maximum/minimum brightness temperature values are set according to the different seasons, and multiple detection results are used to indicate a level of confidence in cloud detection [39]. Because the heat capacity of the land is lower than that of the ocean, especially at night, the second module mainly focuses on cloud detection algorithms over land and ocean surface backgrounds. Cloud identification applies a multispectral test with auxiliary LC and DEM data in many situations such as the obvious differences in the infrared brightness temperature over plateaus. Meanwhile, in near-polar regions, the BTD7.3_11 test can be employed to solve the temperature inversion problem over snow-cover surfaces. The fused values of Groups 1 and 2 are considered cloud if more than four tests are satisfied, clear when fewer than two tests are satisfied, and ambiguous under other conditions (probably cloud and probably clear). The third group of tests is used to correct the cloud and probably cloud results, because they contain unavoidable ambiguities even when certain tests are applied, and the dynamic threshold lookup table (LUT) method is introduced to assess the ambiguity of the obtained cloud detection results mainly at middle and high latitudes. At the same time, considering the brightness temperature characteristics of the varying land elevations, the cloud detection tests and thresholds are adjusted for the plateau areas classified as 2500–3000 m, and above 3000 m. Due to the LST changes at different latitudes in the same longitude range, which makes some detections easily expand the cloud boundary range under the same threshold setting in low-latitude areas. To address this issue, cloud overestimates in low-latitude areas are adjusted to more accurately represent probably clear and clear sky. At the same time, cloud detection algorithms are described for the underlying ocean, land, and polar areas, and the cloud properties can also be used to select different cloud detection algorithms. The test details are provided in Sections III-A–III-D for different situations.

A. Cloud Tests Over Land

The highest uncertain in cloud tests over land is mainly caused by the different surface types with distinct brightness temperatures, and a series of tests are involved mainly focused on Group 2 to solve this problem. Channels 2, 6, and 7 (centered at 3.8, 10.8, and 12.0 μm, respectively) emit thermal radiation, where attenuation primarily occurs by water vapor, increasing in strength with increasing the wavelength [40]; however, the brightness temperature of clear pixels mainly depends on the land surface, clear-sky atmospheric properties,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:44:57 UTC from IEEE Xplore. Restrictions apply.

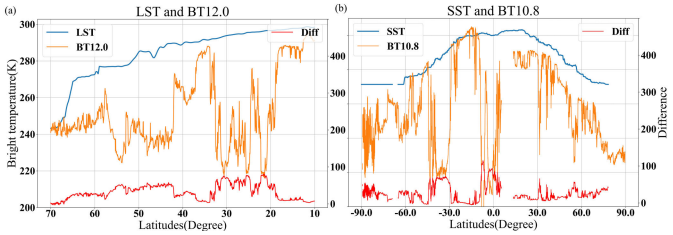


Fig. 2. Cloud tests combined with (a) LST and (b) SST. Zonal LST\SST with blue line, BT12.0\BT10.8 with orange line, and the difference between the LST and BT12.0\SST and BT10.8 with red line at longitude 100°E within the 70°N–10°N\90°N–90°S range on March 15, 2022.

and seasons, which affects the detection tests and thresholds [41]. There is a significant difference in the infrared brightness temperature of the land surface in plateau regions, and the low brightness temperature of the desert surface can lead to cloud misjudgment. The thresholds of BT3.08, BT10.8, and BT12.0 can be adjusted according to the elevation by the DEM data, and the threshold ranges from approximately 251–256 K during different seasons. The threshold of the cloud detection algorithm is further adjusted according to the LC data.

In addition, the cloud detection test combined with LST can better solve the cloud misclassification problem brought by complex land surface brightness temperature differences in the low- and middle-latitude area. Zonal LST and BT12.0 data and the difference between the LST and BT12.0 curves at 100°E longitude between 70° and 10°N latitudes, mainly covering land are chosen as shown in Fig. 2(a).

B. Cloud Tests Over Ocean

Compared to land scenes, the ocean temperature is consistent between day and night in infrared channels [42], but there is a temperature change with latitude variation. The cloud detection algorithm of the MOD35 cloud product combined with ocean temperature data can improve the detection accuracy over the ocean. The 10.8 μm channel is mainly influenced by water vapor absorption in the atmosphere, which varies with the observation angle. The detection error can be solved considering the difference between the SST and BT10.8 detection results by increasing the observation angle and latitude. We chose zonal SST and BT3.8 data and the difference between the SST and BT3.8 curves at 0° longitude between 90°N and 90°S latitudes, mainly covering ocean areas, as shown in Fig. 2(b), This approach is particularly effective considering the lower brightness temperatures of high clouds according to the increased test difference values and improves the accuracy of cloud detection in high-latitude areas with latitude changes.

C. Cloud Test in Polar Regions

Discriminating clear from cloudy conditions in polar regions is more difficult than that in other regions, such as land and ocean, because the snow surface brightness temperature is close to the cloud brightness temperature, which increases the cloud detection difficulty [43]. MOD35 applies BTD7.3_11 to solve the land surface cloud algorithm of ice and snow in

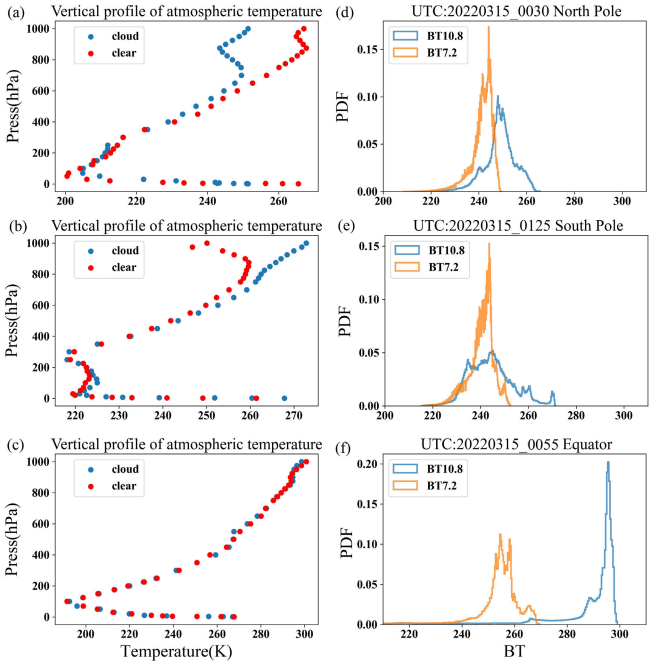


Fig. 3. Vertical profile of the atmospheric temperature from the ERA5 reanalysis dataset in the (a)–(c) Antarctic, Arctic, and equatorial regions, in which the red dots indicate clear-sky pixels, and the blue dots indicate cloud pixels, and (d)–(f) brightness temperature statistics of single-scene images BT7.2 and BT10.8.

polar regions at night. In the case of inversion, BT11 mainly originates from the cloud top, thus clear-sky BTD7.3_11 value will decrease [44]. With the analysis of ERA5 reanalysis data shown in Fig. 3(a)–(c), the vertical profile of the atmospheric temperature from the land surface at 1000 hPa to the top of atmosphere at 0 hPa can be extracted to observe the cloud and channel brightness temperature surface greatly changes. For example, the near-surface snow temperature in Antarctica is approximately 22 K lower than that of clouds. In addition, surface inversion occurs when BT7.2 is greater than BT10.8 in the polar area, and the Antarctic inversion effect is more obvious. The brightness temperature statistics of the two channels are shown in Fig. 3(d)–(f). It can be concluded that the difference between bright temperature of these two channels in polar regions is smaller than that in equatorial regions [45]. Therefore, the relative difference between BT7.2 and BT10.8 can be used to solve the cloud detection error in inversion over ice and snow surface in polar regions.

D. Dynamic Threshold Test

To improve cloud detection, we considered using not only the static threshold method, but also the dynamic threshold method to reduce the detection errors at middle and high latitudes. Studies have shown that the decreased surface brightness temperature caused by the variation in the solar zenith angle (SOZ) can lead to the cloud overextraction phenomenon [23]. The decrease in the surface brightness temperature due to latitude variation can also impact the cloud detection threshold. For example, a dynamic threshold cloud test has been published based on the threshold changes caused

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:44:57 UTC from IEEE Xplore. Restrictions apply.

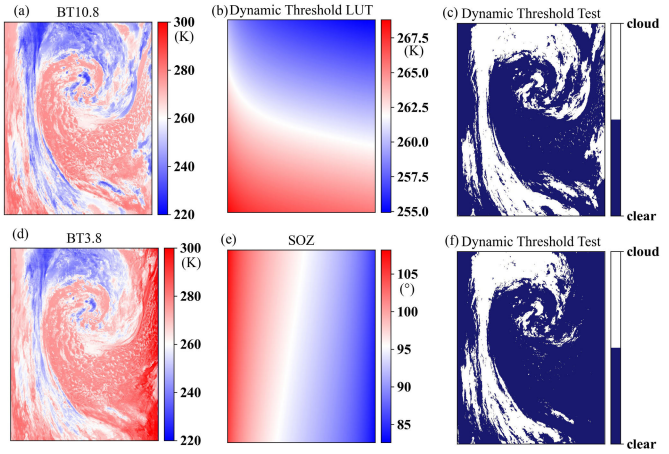


Fig. 4. (a) Brightness temperature of the 10.8 μm channel of FY-3E at UTC 0955 on March 10, 2022. (b) Dynamic threshold LUT of BT10.8. (c) Cloud detection result corresponding to the dynamic threshold LUT of BT10.8. (d) Brightness temperature of the 3.8 μm channel of FY-3E at the same time. (e) SOZ of BT3.8. (f) Cloud detection result corresponding to the dynamic threshold LUT of BT3.8.

by SOZ through the radiation and brightness temperature characteristics of the channel under clear sky conditions by the AVHRR, which improves the accuracy of cloud pollution [46]. Based on the gradual change in BT10.8 with latitude [also refer to the BT10.8 curve in Fig. 2(b)], we can establish a dynamic threshold LUT to solve the problem of cloud misjudgment caused by the low surface temperature in middle and high latitude regions as shown in Fig. 4. In this article, multiple linear regression method is used to construct a dynamic threshold LUT to minimize the residual sum of squares between the observation target and the linear approximation prediction target. The initial threshold for detecting the brightness temperature channel is set considering multiple linear regression, according to the linear regression equation expressed in (1), where X is the latitude matrix of the image, Y is the threshold LUT of the prediction threshold array, θ_i $(I = 0, 1, \ldots, n)$ is the weight vector parameter, ε is the error term, and Fig. 4(c) and (f) shows the results of the dynamic threshold test of BT3.8 and BT10.8, indicating that this method can more effectively detect clouds in BT10.8 than in BT3.8 in middle-latitude regions and that BT10.8 is less influenced by the SOZ

$$Y(x) = \theta_0 + \theta_1 X_1 + \theta_2 X_2 + \theta_3 X_3 + \cdots \theta_n X_n + \varepsilon. \quad (1)$$

E. Different Level Clouds Detection

In addition to considering cloud detection application scenarios, cloud properties are important factors in cloud detection tests. FY-3E provides only infrared brightness temperature channels for detection, and the SOZ in each scene is basically more than 82° at dawn and dusk with less solar illumination. The bi-spectrum technique of the infrared channel can be developed involving the difference between the emittances measured in two channels. The emittance channels, in regard to the upwelling infrared radiation principally influenced by water vapor in the atmosphere, can be applied to clouds at different levels, but the mid\low-level clouds are more different in detection and easily cause detection mistakes.

The BTD8.6-7.2 test is used for mid/low-level clouds detection over the sea, and for the detection of lower-level clouds in regions with low relative humidity in the middle atmosphere [42]. BT8.6 can be used to differentiate between various cloud top phases, while the low-level water vapor has slightly influenced BT7.2. Therefore, this test can better facilitate the detection of mid/low-level clouds.

The BTD11_12 test has been widely used in the cloud detection algorithms of MYD35, Cloud and Surface Parameter Retrieval (CASPR) system, AVHRR Processing Scheme over Clouds, Land, and Ocean (APOLLO) scheme, and CLAVR-1 scheme [47]. BTD10.8_12.0 is employed to identify lower-level water clouds, broken clouds, and thin cirrus clouds over the ocean, and it is especially effective for the detection of stratocumulus clouds along the coast [48]. Moreover, the single channel brightness temperature test is likely to misjudge lower-level water clouds as clear sky, but the BTD10.8_12.0 test can detect low-level clouds with a relatively high brightness temperature in ocean areas.

The BTD11_3.8 test can detect low-level clouds and water clouds at night [49], and it is suitable for the detection of clouds both over the ocean and land, but the difference in the brightness temperature between the two channels is small in desert areas. This algorithm takes advantage of the fact that the low-level clouds in BT3.8 are higher than those in BT10.8, and the total amount of water vapor in the atmosphere under clear-sky conditions is insensitive to BTD10.8_3.8, but is very sensitive to the total water vapor in clouds.

The cloud brightness temperature of the infrared channels is related to the cloud optical thickness, cloud particle radius, and cloud-top temperatures obtained with the cloud radiance model [50]. The BTD3.8_12.0 test can be applied in the determination of thin clouds under various cloud conditions; studies have shown that the maximum difference in the brightness temperature between 3.8 and 12.0 μm is mainly centered at a cloud optical thickness of 1 with decreasing cloud-top temperatures under the different cloud particle radii, and we simulate the theoretical relationship between BTD3.8_12.0 and the cloud optical thickness δ for different cloud particle radii (Rd), three water droplet sizes (5, 10, and 20 μm) [51], and three ice particle sizes (20, 40, and 100 μm) at different press levels (550, 850, 1013.25 hPa), and set the satellite zenith angle (SAZ), SOZ, and relative azimuth angle between sun and satellite (REA) through the RSTAR radiative transfer model [52] as shown in Fig. 5, indicating that this test can effectively detect thin clouds ($\delta = 1$) in low or middle-level clouds and even ice cloud.

IV. Results and Analysis

Through global cloud detection of the above four Groups in several scenes, the global ascending or descending cloud detection results obtained by the early morning orbiting FY-3E satellite during the four seasons from 2021 to 2022 are shown in Fig. 6, and we qualitatively and quantitatively evaluated the detection results. First, the algorithms were evaluated through test examples. Second, cloud verification was verified by published satellite cloud products, including polar-orbiting satellite products, geostationary satellite products, and ground

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:44:57 UTC from IEEE Xplore. Restrictions apply.

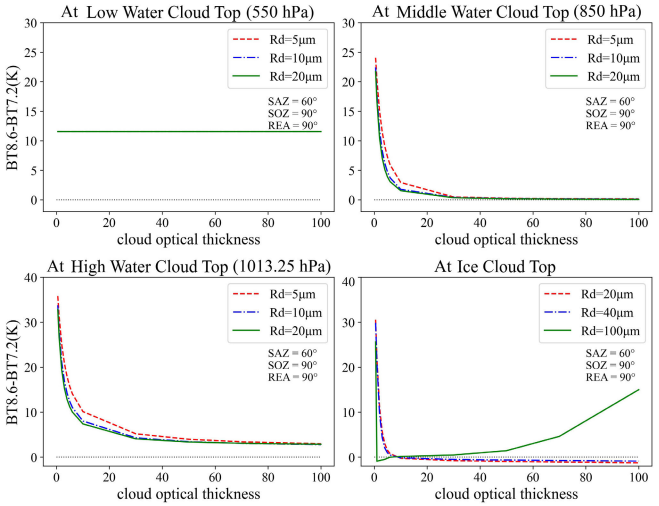


Fig. 5. Theoretical relationship between BTD3.8_12 and the cloud optical thickness for different cloud particle radii (Rd) at SAZ = 60°, SOZ = 90°, and REA = 90° over water and ice cloud tops.

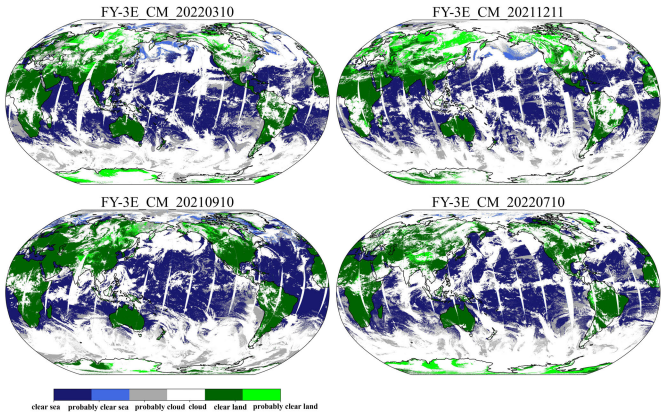


Fig. 6. FY-3E (right) ascending/(left) descending global cloud detection results (CM) in the different seasons (on September 10 and December 11, 2021, and March 10 and July 10, 2022).

site data. Finally, the accuracy of the test results was evaluated.

A. Evaluation of the Algorithms

In regard to qualitative evaluation over land areas, the results of simple tests such as the BT3.8, BT10.8, and BT12.0 tests for the desert land and plateau areas surrounding the Tibetan Plateau are shown in Fig. 7. False-color images can generally display clear cloud distributions within the yellow range. Clouds in the Tibetan Plateau and Taklimakan Desert areas could be accurately detected in false-color cloud images for December, March, and July, supporting the previous test interpretation.

Regarding cloud-type detection over the ocean, Fig. 8 shows the detection of low-level clouds using the BTD10.8_12.0 algorithm. Fig. 8(a) shows the corresponding BT10.8 image, which includes a wide distribution of low-level clouds and broken clouds with relatively high brightness temperatures (approximately 275–285 K) in the Atlantic Ocean off the west coast of South America on July 12, 2021. The detection results in Fig. 8(b) match the

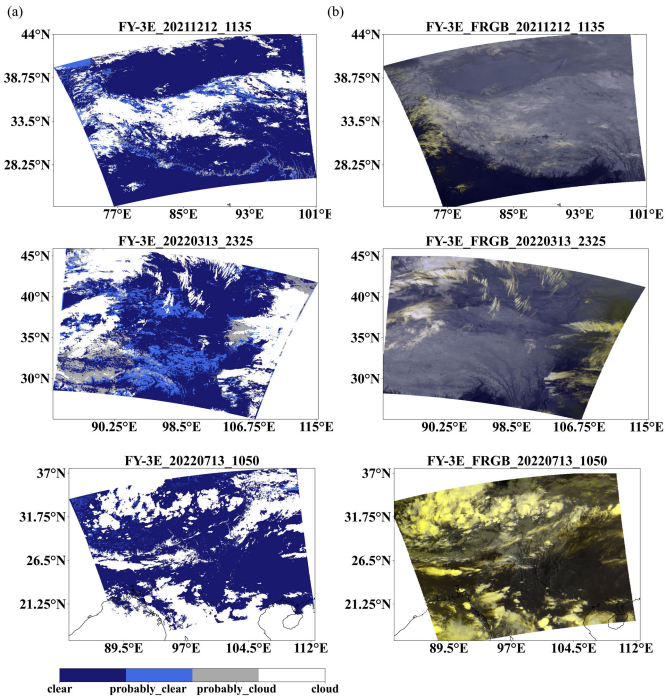


Fig. 7. (a) Cloud detection results in plateau area-plateau desert land type. (b) FY-3E channel stretched to false-color RGB (0.38, 10.8, and 12.0 μm) composite image at the corresponding time, and the yellow-white distribution denotes cloud.

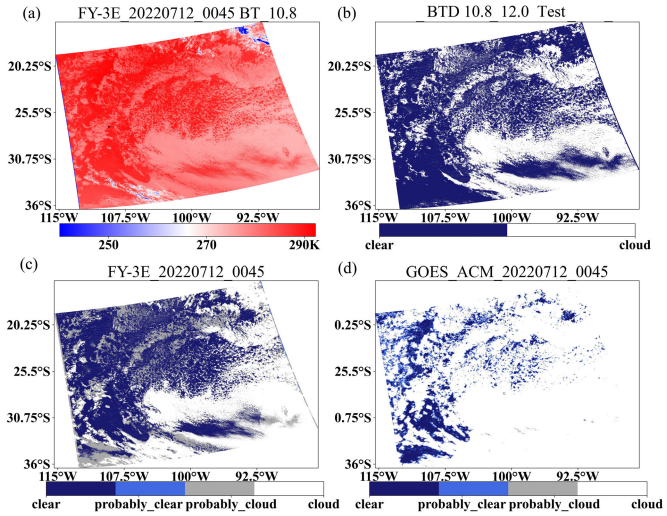


Fig. 8. (a) BT10.8 map (low brightness temperatures are denoted in red, and high brightness temperatures are denoted in deep blue) and UTC: 20220712 0045. (b) BTD10.8_12.0 test results of low-level cloud detection. (c) FY-3E cloud detection results. (d) GOES-16\ACM on July 12, 2022.

cloud distribution in Fig. 8(a) well, and Fig. 8(c) and (d) shows the final cloud results of FY-3E and GOES-16\ACM. In comparison, the probable cloud distribution depicted in Fig. 8(c) indicates that the other tests are not suitable substitutes for the BTD10.8_12.0 test in the identification of thin clouds, especially broken clouds identified as probably clouds, as shown by the cloud distribution in Fig. 8(c), and the difference between the results is obvious, indicating that clouds are not detected in some locations. Based on the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:44:57 UTC from IEEE Xplore. Restrictions apply.

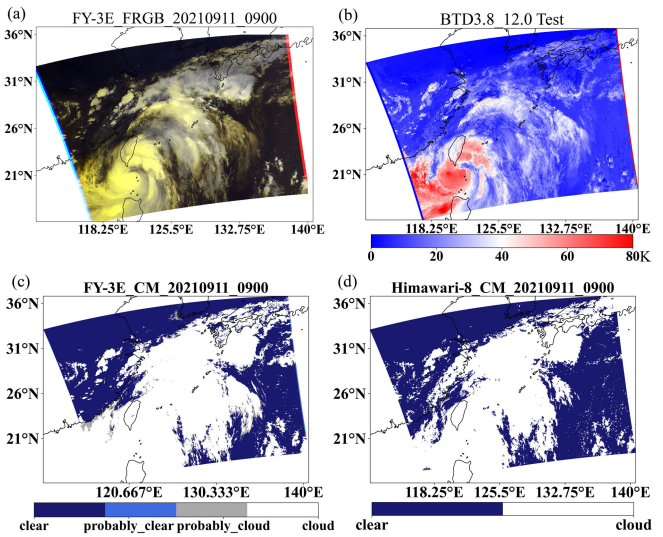


Fig. 9. (a) FY-3E channel stretched to false-color RGB (0.38, 10.8, and 12.0 μm) composite image. (b) BTD3.8_12.0 test in cirrus cloud detection. (c) FY-3E cloud detection result (CM). (d) Himawari-8 cloud production (CM) on September 11, 2021.

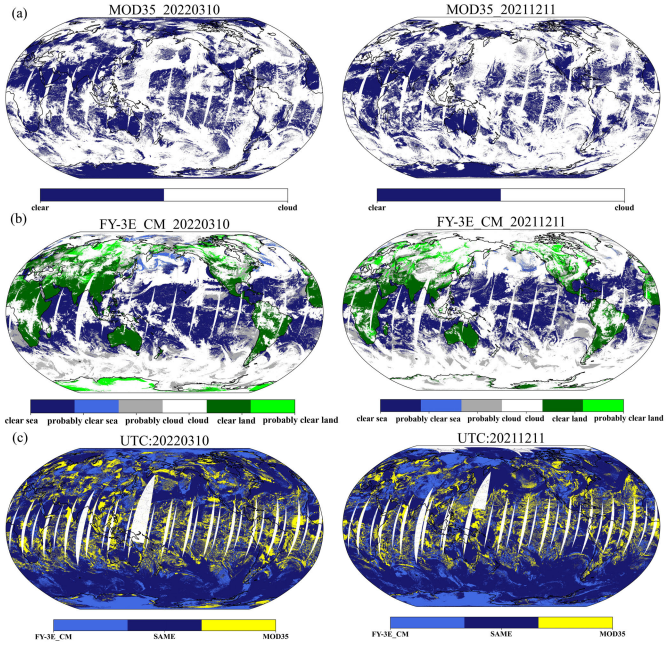


Fig. 10. (a) MOD35 cloud product global composite map on March 10, 2022 and December 10, 2021. (b) FY-3E cloud detection (CM) global composite map. (c) Differences between the MOD35 and FY-3E cloud detection results.

limitations of the FY-3E channel setup, our detection method does not involve transparent thin cloud tests.

In addition, Fig. 9(b) shows one case in the coastal region of China using the BTD3.8_12.0 algorithm. The higher brightness temperature is consistent with the clouds in the false-color image (over the yellow–white area) shown in Fig. 9(a), and Fig. 9(c) and (d) shows the final cloud detection results of the FY-3E and Himawari-8 cloud product. The cloud distribution in Fig. 9(c) reveals that this algorithm can identify cirrus clouds, especially deep convection and cirrus stratus clouds with largely different test values, and the results are even better than those of Himawari-8 with a matched accuracy of 86.21%.

B. Evaluation Based on Polar-Orbiting Satellite Data Cloud Verification

The verification of the FY-3E global cloud detection results based on the MOD35 cloud product is shown in Fig. 10. The differences between these two cloud products are shown. The light blue distribution indicates the FY-3E cloud and MOD35 clear-sky results, the yellow distribution indicates the MOD35 cloud and FY-3E clear-sky results, and the dark blue distribution indicates the same results between the two products. The results showed that the distribution of MOD35 clouds in the equatorial region was larger than that of FY-3E clouds, mainly due to the difference in water cloud extraction. The detection of FY-3E clouds in polar regions, especially in the Antarctic region, exhibited favorable continuity and rationality. However, the two data products suffer high uncertainties due to time-matching errors, showing differences in dynamic clouds between the different polar-orbiting satellites at different observation times, especially for changing broken clouds. Therefore, a three-satellite polar-orbiting network with shorter global observation periods is needed.

C. Evaluation Based on Stationary Satellite Data Cloud Verification

To reduce the verification error of the cloud detection algorithm due to time-matching issues, verification work can be performed involving geostationary satellite data with a high temporal resolution. Himawari-8\AHI and GOES-16\ABI false-color maps were used to assess the temporal and spatial distributions of clouds in areas in the Eastern and Western Hemispheres. Due to the lack of visible light data during some periods, it is difficult to obtain real-time Himawari-8\GOES-16 RGB composited false-color images for comparison. Hover, the 10.8 μm channel, which is sensitive to water vapor absorption can be selected for comparison and verification purposes, and the corresponding brightness temperature can also be used to distinguish cloud types [53]. The results were verified by using the brightness temperature retrieved from Himawari-8\AHI 10.8 μm and GOES-16\ABI 10.4 μm channel images, and the FY-3E whole-orbit results, as captured in mosaic single-scene images, were superimposed on BT10.4 image, as shown in the Figs. 11 and 12. The time-matching error between these two satellites' data were controlled within 30 min, and it could be concluded that the FY-3E cloud results suitably agreed with the cloud distribution characteristics of low bright temperature in the 10.8 μm channel of the Himawari-8/AHI and BT10.4 channel of the GOES-16/ABI. Broken clouds and cirrus clouds distributed in the Atlantic Ocean region in the Western Hemisphere could be effectively identified, and the clouds over land near 60°N in the Eastern Hemisphere could also be suitably performed. The single-track cloud distribution results exhibited favorable continuity with the change in latitude between the two comparison results.

Quantitative verification can be conducted via Himawari-8 24 h cloud products and GOES-16\ACM products covering

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:44:57 UTC from IEEE Xplore. Restrictions apply.

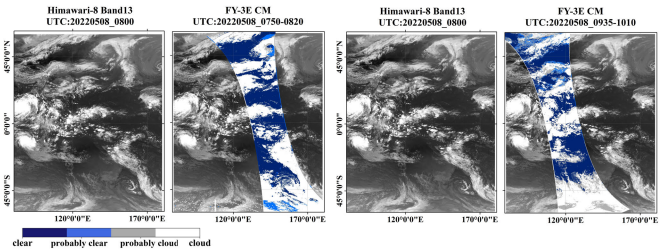


Fig. 11. Comparison of the FY-3E cloud detection results (CM) to the BT10.4 image of Himawari-8\AHI UTC: 20220508.

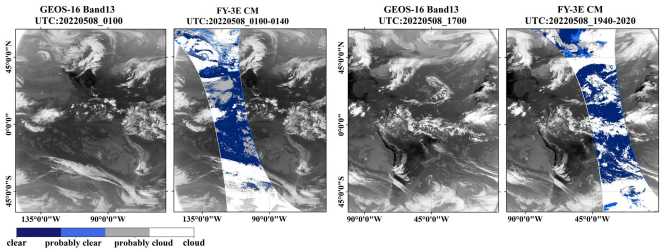


Fig. 12. Comparison of the FY-3E cloud detection results (CM) to the BT10.4 image of GOES-16\ABI UTC: 20220508.

most of the Eastern and Western Hemispheres, respectively. The hit rate (HR), leakage rate (LR), false alarm rate (FAR), and accuracy (ACC) were calculated to examine the accuracy of the detection results with (2)–(6) [54], [55], [56], [57], [14], where FY denotes the FY-3E detection result pixels, SCP denotes the validation satellite cloud product pixels, such as the Himawari-8 cloud product or GOES-16\ACM product, cloud denotes the cloud detection result pixels, clear denotes the clear detection result pixels, TP\TN denotes the number of pixels correctly recognized among the cloud\clear pixels, and FP\FN denotes the error in the recognized cloud\clear pixels.

The verification scope is 60°N–60°S\80°E–180°E in the Eastern Hemisphere and 60°N–60°S\156°W–6°E in the Western Hemisphere, which can be further divided into 60°N–60°S and 30°N–30°S, i.e., low- and medium-latitude intervals, respectively. Approximately 30 scenes encompassing 6–7 track detection results were verified by Himawari-8, and approximately 60 scenes of GOES-16\ACM were used for verification on a daily basis. The time-matching error was approximately 5 min. Moreover, due to the different classification schemes of the results in the cloud products, FY-3E cloud\clear and probable cloud\clear results were compared to SCP cloud\clear results, and the verification results during the four seasons are listed in Tables IV–VII

$$\mathrm{CLOUD_{HR}} = (\mathrm{FY_{clear}}\ \&\ \mathrm{SCP_{clear}})/\mathrm{SCP_{clear}} \quad (2)$$

$$\mathrm{CLEAR_{HR}} = (\mathrm{FY_{cloud}}\ \&\ \mathrm{SCP_{cloud}})/\mathrm{SCP_{cloud}} \quad (3)$$

$$\mathrm{LR} = (\mathrm{FY_{clear}}\ \&\mathrm{SCP_{cloudy}})/N \quad (4)$$

$$\mathrm{FAR} = (\mathrm{FY_{cloud}}\ \&\ \mathrm{SCP_{clear}})/N \quad (5)$$

$$\mathrm{ACC} = (\mathrm{TP}+\mathrm{TN})/(\mathrm{TP}+\mathrm{FN}+\mathrm{FP}+\mathrm{TN}). \quad (6)$$

The verification results in Tables IV and V show that the overall cloud HR within 60°N–60°S\80°E–180°E during the four seasons is 80.53%, the clear HR is 72.35%, the cloud detection LR is below 8.66%, and the FAR is below 12.22%. With the use of auxiliary data, the average ACC of the cloud detection algorithm is enhanced by 3.49%, the cloud HR is improved by over 17.28%, LR is reduced by 7.65%, and FAR is increased by 4.16%.

The cloud detection accuracy at low latitudes (30°N–30°S) is slightly higher than that at middle latitudes (60°N–60°S)

TABLE IV

FY-3E CLOUD DETECTION AND HIMAWARI-8 CARE 24-h CLOUD PRODUCT COMPARISON VERIFICATION WITHIN THE RANGE OF 60°N–60°S\80E°–180°E IN THE DIFFERENT SEASONS

Scenes	60°N-60°S\ 80E°-180°E	Autumn (%)	Winter (%)	Spring (%)	Summer (%)	Average (%)
total	ACC	81.31	78.23	78.44	78.53	**79.13**
	CLOUD_HR	79.33	80.8	78.56	83.43	**80.53**
	CLEAR_HR	75.18	68.8	75.07	70.37	**72.35**
	LR	8.12	9.87	10.11	6.52	**8.66**
	FAR	10.57	11.9	11.44	14.95	**12.22**
ocean	ACC	80.97	79.12	79.22	77.92	**79.31**
	CLOUD_HR	85.53	87.14	85.16	87.04	**86.22**
	CLEAR_HR	68.51	62.88	70.73	66.03	**67.04**
	LR	3.4	4.06	4.04	3.84	**3.84**
	FAR	9.43	10.25	10.19	12.85	**10.68**
land	ACC	79	74.96	75.1	79.16	**77.05**
	CLOUD_HR	65.11	62.48	58.44	64.33	**62.59**
	CLEAR_HR	84.35	74.69	81.5	80.46	**80.25**
	LR	4.71	5.82	6.07	2.68	**4.82**
	FAR	1.14	1.64	1.25	2.1	**1.53**

TABLE V

FY-3E CLOUD DETECTION AND HIMAWARI-8 CARE 24-h CLOUD PRODUCT COMPARISON VERIFICATION WITHIN THE RANGE OF 30°N–30°S\80°E–180°E IN THE DIFFERENT SEASONS

Scenes	30°N-30°S\ 80E°-180°E	Autumn (%)	Winter (%)	Spring (%)	Summer (%)	Average (%)
total	ACC	81.65	78.46	81.78	81.37	**80.82**
	CLOUD_HR	83.04	84.43	80.91	81.62	**82.50**
	CLEAR_HR	75.31	70.9	81.62	75.5	**75.83**
	LR	6.76	6.73	6.93	7.35	**6.94**
	FAR	11.59	14.81	11.29	11.28	**12.24**
ocean	ACC	82.44	79.09	82.84	82.14	**81.63**
	CLOUD_HR	87.3	88.1	87.84	85.96	**87.3**
	CLEAR_HR	72.48	69	79.89	72.83	**73.55**
	LR	3.96	4.18	3.8	4.48	**4.11**
	FAR	10.69	13.63	10.28	10.31	**11.23**
land	ACC	75.46	77.8	72.4	76.77	**75.61**
	CLOUD_HR	64.92	71.78	59.38	61.12	**64.3**
	CLEAR_HR	77.06	74.48	81.77	80.62	**78.48**
	LR	2.79	2.55	3.13	2.87	**2.84**
	FAR	0.9	1.18	1.01	0.98	**1.02**

TABLE VI

FY-3E CLOUD DETECTION AND GOES-16\ACM CLOUD PRODUCT COMPARISON VERIFICATION WITHIN THE RANGE OF 60°N–60°S, 156°W–6°E IN THE DIFFERENT SEASONS

Scenes	60°N-60°S\ 156°W-6°E	Autumn (%)	Winter (%)	Spring (%)	Summer (%)	Average (%)
total	ACC	77.5	76.29	73.37	74.97	**75.53**
	CLOUD_HR	65.35	68.05	66.2	64.16	**65.94**
	CLEAR_HR	78.11	81.73	77.96	84.42	**80.56**
	LR	17.8	20.68	19.85	22.74	**20.26**
	FAR	4.19	3.03	6.78	2.29	**4.07**
ocean	ACC	78.33	77.57	73.61	74.02	**75.88**
	CLOUD_HR	67.5	69.54	67.39	67.3	**67.93**
	CLEAR_HR	74.7	78.42	76.22	77.46	**76.7**
	LR	14.48	16.29	15.58	18.5	**16.21**
	FAR	3.02	1.66	3.84	1.61	**2.53**
land	ACC	78.36	69.72	70.96	72.73	**72.94**
	CLOUD_HR	62.14	53.91	50.46	49.33	**53.96**
	CLEAR_HR	77.64	83.39	83.35	89.28	**83.41**
	LR	3.32	4.38	4.27	4.23	**4.05**
	FAR	1.17	1.37	2.94	0.67	**1.54**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:44:57 UTC from IEEE Xplore. Restrictions apply.

TABLE VII

FY-3E CLOUD DETECTION AND GOES-16\ACM CLOUD PRODUCT COMPARISON VERIFICATION WITHIN THE RANGE OF 30°N–30°S, 156°W–6°E IN THE DIFFERENT SEASONS

Scenes	30°N-30°S\ 156W°-6°E	Autumn (%)	Winter (%)	Spring (%)	Summer (%)	Average (%)
total	ACC	75.84	72.38	70.41	70.24	**72.22**
	CLOUD_HR	57.16	58.28	55.33	53	**55.94**
	CLEAR_HR	89.94	94.21	92.48	93.75	**92.59**
	LR	20.81	26.25	26.11	28.2	**25.34**
	FAR	3.11	1.38	3.48	1.56	**2.38**
ocean	ACC	76.64	72.58	72.64	70.06	**72.98**
	CLOUD_HR	57.45	59.69	57.98	56.44	**57.89**
	CLEAR_HR	86.29	88.67	90.36	89.58	**88.73**
	LR	17.35	20.84	20.74	22.8	**20.43**
	FAR	2.43	1.21	2.95	1.32	**1.98**
land	ACC	79.53	67.88	66.09	73.89	**71.85**
	CLOUD_HR	56.68	46.43	39.2	43.35	**46.42**
	CLEAR_HR	86.58	92.36	95.44	95.33	**92.43**
	LR	3.46	5.4	5.37	5.4	**4.91**
	FAR	0.68	0.16	0.52	0.24	**0.40**

because the brightness temperature fluctuation over the ocean is smaller than that over land with complex surface types in low-latitude regions. The cloud HR is 82.50%, the clear HR is 75.83%, the cloud LR is 6.94%, and the FAR is 12.24%.

The seasonal variation was analyzed to determine the seasonal applicability of the cloud detection algorithm. The cloud/clear HR in summer could reach more than 81%, which is approximately 2%–6% higher than that in spring, and the detection error was mainly distributed in land areas and was largely found in spring (March) at higher latitudes. This occurred because the land surface types are diverse and the seasonal brightness temperature fluctuation is considerable. For example, the low surface temperature in arid and semiarid regions in winter reduces the threshold of the brightness temperature. Another reason is that there is a large snow-covered area on land at 60°N, resulting in the inversion phenomenon. At the same time, low-level clouds could be detected over land due to small detection deviations with a lower proportion of the land area in multiscene images, but the overall detection accuracy was satisfactory, as shown in Fig. 13(a) and (b). The land cloud/clear HR is 90.40%/58.77%, the ocean cloud/clear HR is 92.34%/77.90%, and the total ACC is 83.52%.

The verification index ACC can better reflect the overall detection accuracy of clouds and clear to avoid cloud or clear apparently over-extraction but obtain a high HR. Combined with the FY-3E single-view cloud and clear-sky results for 2021 and 2022, the total ACC, including ocean and land areas, and the land\ocean ACC during the various seasons are shown in Fig. 14. The mean total ACC is over 79%. Due to the complex land surfaces, the average ACC in land areas is 75.61% (30°N–30°S) and notably fluctuates during the different seasons, while the ACC in ocean areas is higher, at approximately 81.63% (30°N–30°S), and remains relatively stable. Because the land types brightness temperature are greatly characteristics with latitude change and seasonal change. For example, the decreasing brightness temperature of the plateau and desert surface can easily lead to cloud misjudgment, which increases the difficulty of setting thresholds and selecting tests on a global scale throughout the four seasons. The total ACC in the low-latitude area is

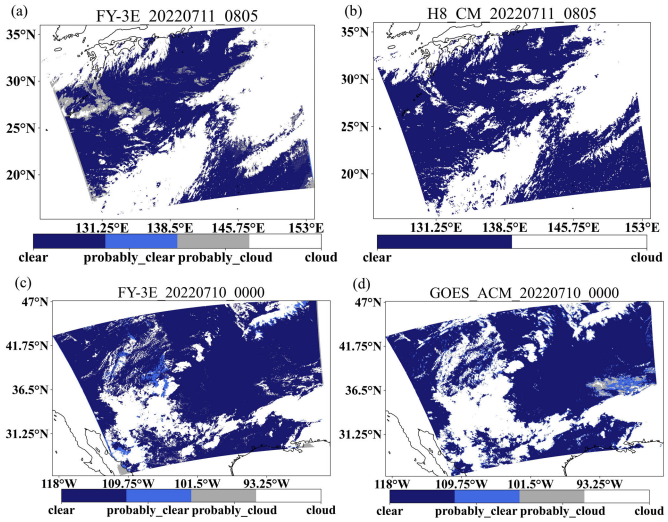


Fig. 13. Comparison and verification of the (a) and (c) FY-3E cloud detection results with the (b) Himawari-8\CARE and (d) GOES-16\ACM cloud products.

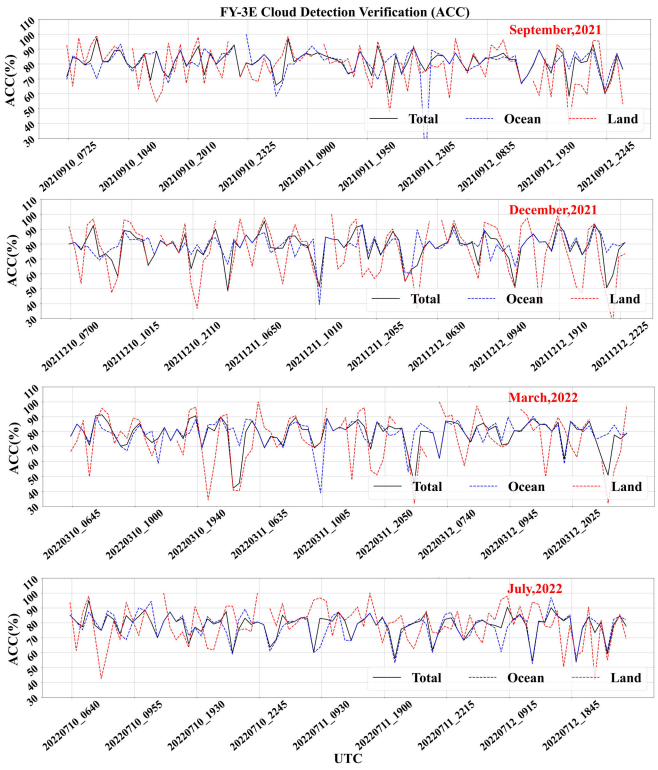


Fig. 14. Daily FY-3E cloud detection verification ACC results of approximately 30 scenes from Himawari-8 cloud production over the ocean and land in September and December 2021 and March and July 2022.

slightly higher than that in the middle-latitude area, and most ACC values exceed 81% (30°N–30°S), except for the value of 78% in December, and the highest ACC can reach 95.00% in summer. At the same time, the lower ACC values are mainly distributed within the range of 30°S–60°S, mainly caused by probably clouds.

The LR of cloud detection is lower than the FAR during the different seasons, manifested as a slight cloud overestimation phenomenon. The LR mainly shows that the cloud results

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:44:57 UTC from IEEE Xplore. Restrictions apply.

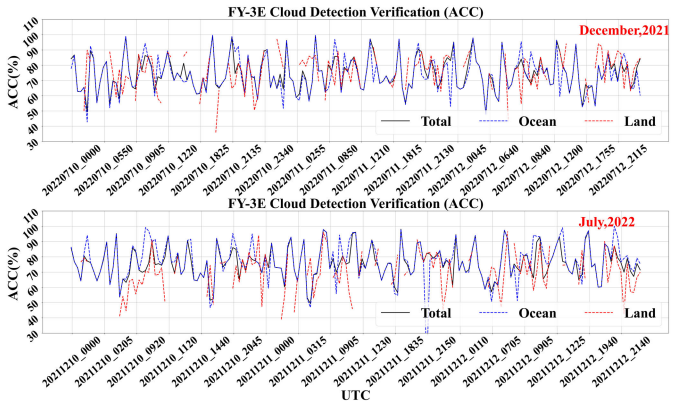


Fig. 15. Daily FY-3E cloud detection verification ACC results of approximately 60 scenes from GOES-16\ACM over the ocean and land in December 2021 and July 2022.

over land areas are lower than those of the Himawari-8 cloud products, and the FAR largely shows that the cloud results over ocean areas are higher than those of the Himawari-8 cloud products. The overall verification accuracy indicates that our proposed FY-3E cloud detection algorithm achieves satisfactory performance in the different seasons and regions at various latitudes.

Validation based on the GOES-16\ACM cloud product is slightly less accurate overall than that based on the Himawari-8 cloud product in the Eastern Hemisphere, as shown in Tables VI and VII. Within the GOES-16\ACM range (60°N–60°S\156°W–6°E), the mean overall cloud HR is 65.94%, the clear HR is 80.56%, the cloud LR is 20.26%, and the FAR is 4.07%. The detection results differ more notably at low latitudes of 30°N–30°S than at latitudes of 60°N–60°S because of the different detection tests for large low-level water clouds at low latitudes. The cloud HR is 55.94%, the clear HR is 92.59%, the cloud LR is 25.34%, and the FAR is 2.38%.

Overall, regarding cloud and clear-sky detection at altitudes of 60°N–60°S, the mean total ACC is above 75.53% during the different seasons, with the highest ACC reaching 77.50% in autumn and the lowest ACC reaching 73.37% in spring. The average ACC in ocean areas is 75.88%, slightly higher than that in land areas (72.94%). This generally agrees with the validation of the ACC by the Himawari-8 cloud products. However, in the 30°N–30°S area, the mean total ACC is 72.22%, and the overall ACC over land is much higher than that over the ocean, which is mainly caused by differences in the test results for thin clouds (even transparent thin clouds), and the daily land\ocean ACC approximately 60 scenes from GOES-16\ACM in winter and summer are shown in Fig. 15, the average one-scene ACC value of ocean is smoother compared to the results of land. The cloud detection accuracy over land is more unstable mainly due to the complexity of land surface types leads to many uncertainties in cloud detection.

The average cloud/clear HR is 65.94%/80.56%, the cloud HR is higher over the ocean than that over land, but the clear HR is higher over land than over the ocean. The cloud LR is 20.26%, which mainly occurs in the ocean region and results in an average FAR of 4.07%, which is smaller over both the ocean and land. It is largely found that the difference in the detection results can be attributed to water clouds and thin clouds in the marine region. Moreover, the clear results of GOES-16\ACM are more conservative, while the cloud results of GOES-16\ACM are mostly beyond those of FY-3E both over the ocean and land.

TABLE VIII

FY-3E CLOUD DETECTION AND GROUND-BASED RADAR\LIDAR DATA COMPARISON VERIFICATION IN THE DIFFERENT SEASONS

Site	Latitude	Longitude	Number	ACC (%)	Cloud_HR(%)	Clear_HR(%)	FAR (%)	LR (%)
ENA	39.09° N	28.03° W	146					
NSA	71.32° N	156.61° W	46	72.58	60.51	84.97	23.87	7.42
SGP	36.61° N	97.49° W	144					

D. Evaluation Based on Ground Site Data Cloud Verification

The ground site data MPLCMASK value-added product can be used for primary verification in middle- and high-latitude areas. We selected cloud masks at three sites during the different seasons for matching verification, which is affected by time-matching issues. In our research, three sites were selected, namely, ENA, NSA, and SGP, updated until 2023, and NSA nearly occurs at the South Pole, thus providing valuable validation data for the polar region. The site attribute of the number of cloud layers >0\=0 is used for comparison verification with cloud detection results for cloud\clear or probably cloud\clear, and the verification accuracy is shown in Table VIII. The experimental results showed that the total ACC is 72.58%, the cloud algorithm slightly underestimates clouds, and the detection error is mainly distributed in single-layer clouds and cloud-top heights at levels from approximately 0–13 km. In addition, the detection error at the NSA site in the Arctic region was relatively high due to the difficulty and challenges of cloud detection over the large area of snow cover on the land surface during winter, as well as the low-temperature inversion phenomenon on the land surface. Another reason is that passive optical sensors are more insensitive to clouds with low extinction or backscatter coefficients than lidar sensors.

V. CONCLUSION AND DISCUSSION

A. Conclusion

In this article, combined with auxiliary data such as SST, LST, and SCM data, a global-scale cloud detection algorithm is developed based on the early morning orbital characteristics and band settings of FY-3E/MERSI-LL. The cloud detection algorithm comprises four detection Groups. The first Group involves a normalized enhancement process of the detected channels, increasing the difference between the cloud and clear brightness temperatures in high-latitude regions and avoiding the image-to-image edge problem. Due to the low thermal capacity of the land, there is a different temperature from the ocean at the same latitude. The second Group mainly entails the implementation of cloud detection algorithms for different surface backgrounds over sea and land, including cloud identification for various situations including plateaus,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:44:57 UTC from IEEE Xplore. Restrictions apply.

highland deserts, and snow cover surfaces. In Group 3, a dynamic threshold LUT approach is used to address the problem of clear misidentification of clouds at high latitudes, and finally, an adjustment for cloud overestimation at low and high latitudes is used in the last Group. The study results reveal the following.

The temporal-spatial comparison of the FY-3E cloud detection results with the global cloud products of the polar-orbiting satellite MOD35 demonstrates that there are high uncertainties associated with the time-matching errors in the two data products and a three polar-orbiting-satellite network with a shorter global observation period is urgently needed. Moreover, the FY-3E cloud detection results suitably agree with the low brightness temperature of the cloud distribution captured in the geostationary satellite Himawari-8\AHI BT10.8 μm and GOES-16\ABI BT10.4 μm images. Broken and cirrus clouds over the ocean in the Western Hemisphere can be effectively identified, clouds can also be accurately determined near 60°N over land areas in the Eastern Hemisphere, and the single-track cloud detection results exhibit favorable continuity with latitude change.

The verification results obtained with the Himawari-8 cloud products show that the overall cloud and clear ACC is 79.13% between 60°N–60°S and 80°E–180°E during all four seasons, the cloud HR is 80.53%, the clear RH is 72.35%, the cloud LR is below 8.66%, and the FAR is below 12.22%. The cloud detection accuracy is slightly higher at low latitudes (30°N–30°S) than at middle latitudes (60°N–60°S), and the detection accuracy remains more stable and is higher in ocean areas than in land areas. Analysis of the seasonal variation reveals that the detection accuracy is higher in summer and autumn than in winter and spring. The overall validation accuracy demonstrates that the detection algorithm performs well in different seasons and latitudes.

Moreover, validation based on the GOES-16\ACM cloud product is slightly less accurate overall than that based on the Himawari-8 cloud product. The overall cloud HR is 65.94%, the clear HR is 80.56%, the cloud LR is 20.26%, and the FAR is 4.07% from 60°N to 60°S. The cloud and clear mean ACC is approximately 75.53% over the different seasons. The FY-3E cloud algorithm achieves a better match than the GOES-16\ACM product, but there is a tendency to underestimate the cloud distribution over the ocean and land in the Western Hemisphere.

Ground site MPLCMASK products, as effective synergistic cloud products with high spatial and temporal observation capacities, are also used to verify the FY-3E cloud results in middle- and high-latitude areas. The experimental results show that the total ACC is 72.58% among three sites (336 times), the cloud algorithm slightly underestimates clouds, and the detection error is mainly distributed in single-layer clouds. At the same time, the detection accuracy in the Arctic region is relatively low, which is also the difficulty and challenge of FY-3E cloud detection during early morning orbit in the polar regions.

B. Discussion

The Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observation (CALIPSO) data can help to further verify and improve the cloud detection algorithm. However, due to the large time-matching error between the FY-3E early morning orbit start and CALIPSO lidar data (the matching range of CALIPSO data is basically 70°N\S–90°N\S), we did not use CALIPSO Vertical Feature Mask (VFM) data for accuracy verification. In future work, we can consider using CALIPSO data to perform verification experiments in polar areas. Moreover, ground-based radar/lidar data are a good option to distinguish atmospheric cloud properties, and they can also be used as the basis for subsequent algorithm improvement through a large number of experiments.

In the validation of the accuracy of the cloud detection algorithm, we selected the geostationary satellite Himawari-8 CARE 24 h cloud product in 15 min intervals and the GOES-16\ACM product in 10 min intervals to ensure a higher accuracy of spatial-temporal matching, but evaluation of the results yielded different conclusions. For example, Fig. 13(c) and (d) shows clouds with leakage areas, which is contrary to the verification results of the Himawari-8 cloud products in the Eastern Hemisphere. This also indicates that there are small differences between the cloud products in terms of coverage region and cloud detection algorithm, as well as the uncertainty introduced by the observed tracks, which should be investigated further.

The cloud detection quality label is an important aspect of the FY-3E cloud detection algorithm update and subsequent quality maintenance. The method for setting the cloud detection quality label is based on the MODIS cloud production [42]. For each threshold detection, a high confident cloud threshold (α) and a high confident clear threshold (γ) are determined, the confidence level of each test is denoted as $G_i(\alpha, \gamma)$, and the calculation confidence level label Q is defined as (7) to represent confident clear ($0.99 \leq Q$), probably clear ($0.95 \leq Q < 0.99$), probably cloudy ($0.66 \leq Q < 0.95$), and cloudy ($0.66 < Q < 0$). The test cloud detection results are combined to output high (cloud\clear), relatively high (cloud\clear), and uncertain labels

$$Q = \sqrt[N]{\prod_{i=1}^{N} G_i}. \tag{7}$$

ACKNOWLEDGMENT

The authors are indebted to the National Meteorological Satellite Center for providing the study data, and they are also grateful to the Open CLASTR project for using the System for Transfer of Atmospheric Radiation (RSTAR) package in this research experiment.

REFERENCES

[1] H. Letu, T. M. Nagao, T. Y. Nakajima, and Y. Matsumae, "Method for validating cloud mask obtained from satellite measurements using ground-based sky camera," *Appl. Opt.*, vol. 53, no. 31, pp. 7523–7533, Nov. 2014, doi: 10.1364/AO.53.007523.

[2] N. G. Loeb et al., "Toward optimal closure of the Earth's top-of-atmosphere radiation budget," *J. Climate*, vol. 22, no. 3, pp. 748–766, Feb. 2009.

[3] G. Tana et al., "Retrieval of cloud microphysical properties from Himawari-8/AHI infrared channels and its application in surface shortwave downward radiation estimation in the sun glint region," *Remote Sens. Environ.*, vol. 290, May 2023, Art. no. 113548, doi: 10.1016/j.rse.2023.113548.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:44:57 UTC from IEEE Xplore. Restrictions apply.

[4] R. A. Schiffer and W. B. Rossow, "The international satellite cloud climatology project (ISCCP): The first project of the world climate research programme," *Bull. Amer. Meteorological Soc.*, vol. 64, no. 7, pp. 779–784, Jul. 1983.
[5] W. B. Rossow and R. A. Schiffer, "Advances in understanding clouds from ISCCP," *Bull. Amer. Meteorological Soc.*, vol. 80, no. 11, pp. 2261–2287, Nov. 1999.
[6] G. Cloud System Science Team, "The GEWEX cloud system study (GCSS)," *Bull. Amer. Meteorological Soc.*, vol. 74, no. 3, pp. 387–399, Mar. 1993.
[7] P. Forster et al., "The Earth's energy budget, climate feedbacks, and climate sensitivity," in *Climate Change 2021: The Physical Science Basis* (Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change), V. Masson-Delmotte, Eds. Cambridge, U.K.: Cambridge Univ. Press, 2021, pp. 923–1054, doi: 10.1017/9781009157896.009.
[8] W. B. Rossow, "Climate data record (CDR) program: Climate algorithm theoretical basis document (C-ATBD)," CDR Program Library, NOAA, U.S., CDR Program Document CDRP-ATBD-0872, 2022. [Online]. Available: https://www.ncei.noaa.gov/pub/data/sds/cdr/CDRs/Cloud_Properties-ISCCP/AlgorithmDescription_01B-29.pdf
[9] R. D. Cess et al., "Interpretation of cloud-climate feedback as produced by 14 atmospheric general circulation models," *Science*, vol. 245, no. 4917, pp. 513–516, 1989, doi: 10.1126/science.245.4917.513.
[10] S. A. Ackerman, K. I. Strabala, W. P. Menzel, R. A. Frey, C. C. Moeller, and L. E. Gumley, "Discriminating clear sky from clouds with MODIS," *J. Geophys. Res., Atmos.*, vol. 103, no. D24, pp. 32141–32157, Dec. 1998, doi: 10.1029/1998JD200032.
[11] R. F. S. Ackerman, K. Strabala, Y. Liu, L. Gumley, B. Baum, and P. Menzel, "Discriminating clear-sky from cloud with MODIS algorithm theoretical basis document (MOD35)," NASA, Version 6.1, 2010. [Online]. Available: https://atmosphere-imager.gsfc.nasa.gov/sites/default/files/ModAtmo/MOD35_ATBD_Collection6_0.pdf
[12] B. A. Baum et al., "MODIS cloud-top property refinements for collection 6," *J. Appl. Meteorol. Climatol.*, vol. 51, no. 6, pp. 1145–1163, Jun. 2012.
[13] T. Wilson et al., "Development and implementation of an electronic crosstalk correction for bands 27–30 in Terra MODIS collection 6," *Remote Sens.*, vol. 9, no. 6, p. 569, Jun. 2017.
[14] P. Minnis et al., "Cloud detection in nonpolar regions for CERES using TRMM VIRS and Terra and Aqua MODIS data," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 11, pp. 3857–3884, Nov. 2008, doi: 10.1109/TGRS.2008.2001351.
[15] P. Minnis et al., "CERES MODIS cloud product retrievals for edition 4—Part I: Algorithm changes," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 2744–2780, 2021, doi: 10.1109/tgrs.2020.3008866.
[16] Q. Z. Trepte et al., "Global cloud detection for CERES edition 4 using Terra and Aqua MODIS data," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9410–9449, Nov. 2019, doi: 10.1109/TGRS.2019.2926620.
[17] L. L. Stowe et al., "Global distribution of cloud cover derived from NOAA/AVHRR operational satellite data," *Adv. Space Res.*, vol. 11, no. 3, pp. 51–54, 1991, doi: 10.1016/0273-1177(91)90402-6.
[18] R. W. Saunders and K. T. Kriebel, "An improved method for detecting clear sky and cloudy radiances from AVHRR data," *Int. J. Remote Sens.*, vol. 9, no. 1, pp. 123–150, Jan. 1988, doi: 10.1080/01431168808954841.
[19] W. L. Smith and E. E. Ebert, "The analysis of polar clouds from AVHRR satellite data using pattern recognition techniques," NASA, NASA Contractor Rep. NASA CR-183300, 1990. [Online]. Available: https://ntrs.nasa.gov/api/citations/19900013552/downloads/19900013552.pdf?attachment=true
[20] L. L. Stowe et al., "AVHRR clear-sky radiation data sets at NOAA/NESDIS," *Adv. Space Res.*, vol. 14, no. 1, pp. 113–116, 1994, doi: 10.1016/0273-1177(94)90358-1.
[21] H. Ishida, T. Y. Nakjima, T. Yokota, N. Kikuchi, and H. Watanabe, "Investigation of GOSAT TANSO-CAI cloud screening ability through an intersatellite comparison," *J. Appl. Meteorol. Climatol.*, vol. 50, no. 7, pp. 1571–1586, Jul. 2011.
[22] H. Ishida and T. Y. Nakajima, "Development of an unbiased cloud detection algorithm for a spaceborne multispectral imager," *J. Geophys. Res.*, vol. 114, no. D7, 2009, Art. no. 2008JD010710, doi: 10.1029/2008jd010710.
[23] H. Shang, L. Chen, H. Letu, M. Zhao, S. Li, and S. Bao, "Development of a daytime cloud and haze detection algorithm for Himawari-8 satellite measurements over central and Eastern China," *J. Geophys. Res., Atmos.*, vol. 122, no. 6, pp. 3528–3543, Mar. 2017, doi: 10.1002/2016jd025659.
[24] C. Liu et al., "A machine learning-based cloud detection algorithm for the Himawari-8 spectral image," *Adv. Atmos. Sci.*, vol. 39, no. 12, pp. 1994–2007, Dec. 2022, doi: 10.1007/s00376-021-0366-x.
[25] A. Heidinger and W. C. Straka, III, "Algorithm theoretical basis document abi cloud mask," NOAA NESDIS Center Satell. Appl. Res., 2012. [Online]. Available: https://www.star.nesdis.noaa.gov/goesr/docs/ATBD/Cloud_Mask.pdf
[26] A. J. Schreiner, T. J. Schmit, and W. P. Menzel, "Observations and trends of clouds based on GOES sounder data," *J. Geophys. Res., Atmos.*, vol. 106, no. D17, pp. 20349–20363, Sep. 2001.
[27] J. Hocking, P. N. Francis, and R. Saunders, "Cloud detection in Meteosat Second Generation imagery at the met office," *Meteorological Appl.*, vol. 18, no. 3, pp. 307–323, Sep. 2011, doi: 10.1002/met.239.
[28] P. Stammes, M. Sneep, J. F. de Haan, J. P. Veefkind, P. Wang, and P. F. Levelt, "Effective cloud fractions from the ozone monitoring instrument: Theoretical framework and validation," *J. Geophys. Res.*, vol. 113, no. D16, 2008, doi: 10.1029/2007jd008820.
[29] H.-J. Lutz, J. B. Gustafsson, and R. Valenzuela-Leyenda, "Scenes and cloud analysis from Meteosat Second Generation (MSG) observations," in *Proc. EUMETSAT Meteorol. Satell. Conf.*, Weimar, Germany, 2003.
[30] E. D. Tomaso and N. Bormann, "Assimilation of ATOVS radiances at ECMWF: First year EUMETSAT fellowship report," EUMETSAT/ECMWF Fellowship Programme, EUMET-SAT, Darmstadt, Germany, Res. Rep. 22, 2011. [Online]. Available: https://www.ecmwf.int/en/elibrary/74220-assimilation-atovs-radiances-ecmwf-first-year-eumetsat-fellowship-report
[31] S. Joo, J. Eyre, and R. Marriott, "The impact of MetOp and other satellite data within the met office global NWP system using an adjoint-based sensitivity method," *Monthly Weather Rev.*, vol. 141, no. 10, pp. 3331–3342, Oct. 2013, doi: 10.1175/mwr-d-12-00232.1.
[32] P. Zhang et al., "FY-3E: The first operational meteorological satellite mission in an early morning orbit," *Adv. Atmos. Sci.*, vol. 39, no. 1, pp. 1–8, Jan. 2022, doi: 10.1007/s00376-021-1304-7.
[33] K. D. Hutchison and J. M. Jackson, "Cloud detection over desert regions using the 412 nanometer MODIS channel," *Geophys. Res. Lett.*, vol. 30, no. 23, p. 2187, 2003.
[34] D. K. Hall, G. A. Riggs, and V. V. Salomonson, "Development of methods for mapping global snow cover using moderate resolution imaging spectroradiometer data," *Remote Sens. Environ.*, vol. 54, no. 2, pp. 127–140, Nov. 1995, doi: 10.1016/0034-4257(95)00137-P.
[35] X. Ri et al., "Cloud, atmospheric radiation and renewal energy application (CARE) version 1.0 cloud top property product from Himawari-8/AHI: Algorithm development and preliminary validation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4108011, doi: 10.1109/TGRS.2022.3172228.
[36] C. Flynn, A. Mendoza, Y. Zheng, and S. Mathur, "Novel polarization-sensitive micropulse LiDAR measurement technique," *Opt. Exp.*, vol. 15, pp. 2785–2790, Apr. 2007, doi: 10.1364/OE.15.002785.
[37] J. Campbell et al., "Full-time, eye-safe cloud and aerosol LiDAR observation at atmospheric radiation measurement program sites: Instruments and data processing," *J. Atmos. Ocean. Technol.*, vol. 19, pp. 431–442, Apr. 2002, doi: 10.1175/1520-0426(2002)019<0431:FTESCA>2.0.CO;2.
[38] K. D. Hutchison, B. D. Iisager, and B. Hauss, "The use of global synthetic data for pre-launch tuning of the VIIRS cloud mask algorithm," *Int. J. Remote Sens.*, vol. 33, no. 5, pp. 1400–1423, Mar. 2012, doi: 10.1080/01431161.2011.571299.
[39] S. Hioki, J. Riedi, and M. S. Djellali, "A study of polarimetric error induced by satellite motion: Application to the 3MI and similar sensors," *Atmos. Meas. Techn.*, vol. 14, no. 3, pp. 1801–1816, Mar. 2021, doi: 10.5194/amt-14-1801-2021.
[40] L. L. Stowe, P. A. Davis, and E. P. McClain, "Scientific basis and initial evaluation of the CLAVR-1 global clear/cloud classification algorithm for the Advanced Very High Resolution Radiometer," *J. Atmos. Ocean. Technol.*, vol. 16, no. 6, pp. 656–681, Jun. 1999.
[41] S. Vemury, L. Stowe, and V. Anne, "AVHRR pixel level clear-sky classification using dynamic threshold (CLAVR-3)," *J. Atmos. Ocean. Technol.*, vol. 18, pp. 169–186, Feb. 2010, doi: 10.1175/1520-0426(2001)018<0169:APLCSC>2.0.CO;2.
[42] R. A. Frey et al., "Cloud detection with MODIS. Part I: Improvements in the MODIS cloud mask for collection 5," *J. Atmos. Ocean. Technol.*, vol. 25, no. 7, pp. 1057–1072, Jul. 2008, doi: 10.1175/2008jtecha1052.1.
[43] W. L. Smith and E. Ebert, "The analysis of polar clouds from AVHRR satellite data using pattern recognition techniques," Space Sci. Eng. Center, Univ. Wisconsin-Madison, Madison, WI, USA, Tech. Rep. 19900013552, 1990.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:44:57 UTC from IEEE Xplore. Restrictions apply.

[44] Y. Liu, J. R. Key, R. A. Frey, S. A. Ackerman, and W. P. Menzel, "Nighttime polar cloud detection with MODIS," *Remote Sens. Environ.*, vol. 92, no. 2, pp. 181–194, Aug. 2004.
[45] W. B. Rossow and L. C. Garder, "Cloud detection using satellite measurements of infrared and visible radiances for ISCCP," *J. Climate*, vol. 6, no. 12, pp. 2341–2369, Dec. 1993.
[46] S. Vemury, L. L. Stowe, and V. R. Anne, "AVHRR pixel level clear-sky classification using dynamic thresholds (CLAVR-3)," *J. Atmos. Ocean. Technol.*, vol. 18, no. 2, pp. 169–186, 1999.
[47] I. Toshiro, "On the temperature and effective emissivity determination of semi-transparent cirrus clouds by bi-spectral measurements in the 10μm window region," *J. Meteorological Soc. Japan. Ser. II*, vol. 63, no. 1, pp. 88–99, 1985.
[48] A. Slingo and H. M. Schrecker, "On the shortwave radiative properties of stratiform water clouds," *Quart. J. Roy. Meteorological Soc.*, vol. 108, no. 456, pp. 407–426, 2010.
[49] M. Derrien et al., "Automatic cloud detection applied to NOAA-11 /AVHRR imagery," *Remote Sens. Environ.*, vol. 46, no. 3, pp. 246–267, Dec. 1993.
[50] L. L. Stowe, "Evaluation of Nimbus 7 THIR/CLE and air force three-dimensional nephanalysis estimates of cloud amount," *J. Geophys. Res., Atmos.*, vol. 89, no. D4, pp. 5370–5380, Jun. 1984.
[51] J. Wang and W. B. Rossow, "Determination of cloud vertical structure from upper-air observations," *J. Appl. Meteorol. Climatol.*, vol. 34, no. 10, pp. 2243–2258, Oct. 1995, doi: 10.1175/1520-0450(1995)034<2243:DOCVSF>2.0.CO;2.
[52] T. Nakajima and M. Tanaka, "Algorithms for radiative intensity calculations in moderately thick atmospheres using a truncation approximation," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 40, pp. 51–69, Jul. 1988, doi: 10.1016/0022-4073(88)90031-3.
[53] J. Key and R. G. Barry, "Cloud cover analysis with Arctic AVHRR data: 1. Cloud detection," *J. Geophys. Res.*, vol. 94, no. D15, pp. 18521–18535, Dec. 1989, doi: 10.1029/JD094iD15p18521.
[54] T. J. Kopp et al., "The VIIRS cloud mask: Progress in the first year of S-NPP toward a common cloud detection scheme," *J. Geophys. Res., Atmos.*, vol. 119, no. 5, pp. 2441–2456, Mar. 2014.
[55] S. Shalev-Shwartz and S. Ben-David, *Understanding Machine Learning: From Theory to Algorithms*. Cambridge, U.K.: Cambridge Univ. Press, 2014.
[56] H. Shang et al., "Diurnal haze variations over the North China plain using measurements from Himawari-8/AHI," *Atmos. Environ.*, vol. 210, pp. 100–109, Aug. 2019, doi: 10.1016/j.atmosenv.2019.04.036.
[57] X. Wang, M. Min, F. Wang, J. Guo, B. Li, and S. Tang, "Intercomparisons of cloud mask products among Fengyun-4A, Himawari-8, and MODIS," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8827–8839, Nov. 2019, doi: 10.1109/TGRS.2019.2923247.



Ni An received the M.S. degree in cartography and geographic information system from Inner Mongolia Normal University, Hohhot, China, in 2015, where she is currently pursuing the Ph.D. degree in geography.

Since 2016, she has been with the College of Resources and Environment, Baotou Teachers' College, Baotou, China. She is also a Guest Student with the State Key Laboratory of the Science and Remote Sensing, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. Her research interests include atmospheric remote sensing and cloud detection.



Huazhe Shang received the B.S. degree in remote sensing from Wuhan University, Wuhan, China, in 2011, and the Ph.D. degree in cartography and geographic information system from the University of Chinese Academy of Sciences, Beijing, China, in 2016. His Ph.D. degree training included the retrieval of cloud droplet size distribution using multiangle and polarization measurements from the POLarization and Directionality of Earth Reflectances (POLDER).

Since 2016, he has been a Research Assistant with the State Key Laboratory of Remote Sensing Science, Chinese Academy of Sciences, Beijing. He was a Visiting Scholar conducting research on cloud retrieval for the multiviewing, multichannel, and multipolarization imager (3MI) with Laboratoire d'Optique Atmosphérique, Université de Lille, Lille, France, in 2019. His research interests include the detection of clouds and recognition of cloud phase information using measurements from geostationary satellites and retrieval of liquid cloud properties from POLDER, 3MI, and directional polarimetric camera (DPC).



Wei Lesi received the B.S. and M.S. degrees from the Physics and Electronic Information College, Inner Mongolia Normal University, Hohhot, China, in 2013 and 2017, respectively.

He is a Research Assistant of sciences with the State Key Laboratory of the Science and Remote Sensing, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include satellite cloud remote sensing and radiative transfer mode.



Xu Ri received the B.S. degree in physics and the M.S. degree in cartography and geographic information system from Inner Mongolia Normal University, Hohhot, China, in 2013 and 2017, respectively. She is currently pursuing the Ph.D. degree in cartography and geographic information system with Northwest Normal University, Lanzhou, China.

She is with the State Key Laboratory of the Science and Remote Sensing, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, as a Guest Student. Her research interests include cloud remote sensing and radiative transfer model.



Chong Shi received the M.S. degree in environmental science from the Institute of Earth Environment, Chinese Academy of Sciences (CAS), Beijing, China, in 2012, and the Ph.D. degree in atmospheric physics from the Institute of Atmospheric Physics, CAS, in 2015.

He was with the Earth Observation Research Center (EORC), Japan Aerospace Exploration Agency (JAXA), Tsukuba, Japan, from 2015 to 2020. He has been with the Center for Global Environmental Research, National Institute for Environmental Studies, Japan, since 2020. He is involved in the algorithm development and validation of aerosol product for GOSAT-2/CAI-2 satellite missions. His research interests include radiative transfer in the atmosphere-ocean systems and remote sensing of atmospheric aerosol and ocean color.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:44:57 UTC from IEEE Xplore. Restrictions apply.



Gegen Tana received the M.S. degree from the Graduate School of Science and Technology, Chiba University, Chiba, Japan, in 2008, and the Ph.D. degree from the Department of Earth Sciences, Chiba University, in 2013.

From 2017 to 2021, she was a Research Assistant with Beijing Normal University, Beijing, China. Since 2022, she has been a Research Assistant with the National Space Science Center, Chinese Academy of Sciences, Beijing. Her research interests include the retrieval of cloud water contents and surface radiation budget with remote sensing technology.



Yuhai Bao received the M.S. degree in physical geography from the Department of Geography, Northeast Normal University, Changchun, China, in 1992, and the Ph.D. degree from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, China, in 1999.

He is a Professor and the Dean with the Department of Geographical Sciences, Inner Mongolia Normal University, Hohhot, China, and the Director of the Key Laboratory of Remote Sensing and Geographic Information System, Inner Mongolia, China. His research interests include remote sensing of resources and environment and natural disasters on the Mongolian Plateau.



Zhaojun Zheng received the B.S. degree in atmospheric science from Nanjing University, Nanjing, China, in 2000, and the M.S. degree in atmospheric physics and atmospheric environment from Peking University, Beijing, China, in 2008.

He has been working at the National Satellite Meteorological Center, Satellite Meteorological Institute, China Meteorological Administration, Beijing, since 2000. His research interests include cryosphere remote sensing, cryospheric change analysis, and optical cloud mask.

Mr. Zheng is a member of WMO EC-PHORS. He was awarded the Research Associate in 2009.



Na Xu received the B.S. degree in atmospheric science from the Ocean University of China, Qingdao, China, in 2005, and the Ph.D. degree in atmospheric physics and environment from the Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, China, in 2010.

She is a Professor with the National Satellite and Meteorological Center, China Meteorological Administration, Beijing. Her research interests include calibration and validation for optical sensors and remote sensing of cloud microphysics.

Dr. Xu is a member of the Global Space-Based Inter-Calibration System (GSICS/WMO). She is an Instrument Scientist of the Medium Resolution Spectral Imager onboard the Chinese FengYun-3D (FY-3D) and an In-Charge of the Calibration and Validation for MERSI II. She is the Chief Director of the Product Generation System (PGS) for FengYun-3 and FengYun-4 Projects.



Lin Chen received the Ph.D. degree from the Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, China, in 2009.

Currently, he works as a Professor with the National Satellite Meteorology Center of China, China Meteorological Administration, Beijing. He has authored or coauthored over 60 science articles in peered-reviewed journals. His research interests include sensor development and design on the polar orbit satellite and cal/val activities of meteorology satellites and aerosols properties retrieval as well as their climate effect by long-term historical satellite data.

Dr. Chen is a Vice Chief Engineer of the FY-3 Rainfall Mission Satellite Ground Segment.



Peng Zhang (Fellow, IEEE) received the M.S. degree in atmospheric optics from the Anhui Institute of Optics and Fine Mechanics, Chinese Academy of Sciences (AIOFM/CAS), Beijing, China, in 1995, and the Ph.D. degree in atmospheric physics from the Institute of Atmospheric Physics, Chinese Academy of Sciences (IAP/CAS), Beijing, in 1998.

He has been the Deputy Director–General of the National Satellite Meteorological Center (NSMC/CMA), Beijing, since 2013. He has intensive experience in conceiving, developing, and operating the FY-3 satellite ground segment. His research interests include atmospheric remote sensing, satellite data assimilation, satellite calibration, and validation.

Dr. Zhang has been the Chief Director of the FY-3 Ground Segment and Chinese TanSat Satellite Ground Segment since 2013 and 2015, respectively. He was the Chair of the Global Space Inter-Calibration System (GSICS) Executive Panel from 2014 to 2017.



Lingmeng Ye received the B.S. degree in geographic information science from the University of Posts and Telecommunications, Nanjing, China, in 2017, and the M.S. degree in surveying and mapping engineering from Southwest Jiaotong University, Chengdu, China, in 2020.

She has been with Huayun ShineTek Technologies Company Ltd., as a Software Research and Development Engineer, Beijing, China, since 2021. Her research interests include cloud detection.



Husi Letu (Senior Member, IEEE) received the B.S. and M.S. degrees in geography from Inner Mongolia Normal University, Hohhot, China, in 1999 and 2002, respectively, and the Ph.D. degree in geosciences and remote sensing from the Center for Environmental Remote Sensing, Chiba University, Chiba, Japan, in 2010.

He was with the Research and Information Center, Tokai University, Tokyo, Japan. He is currently a Professor with the State Key Laboratory of the Science and Remote Sensing, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. He is involved in the algorithm development and validation of the ice cloud product for JAXA's GCOM-C and Himawari-8 satellite missions. His research interests include atmospheric radiative transfer simulation, light scattering calculation, cloud remote sensing, and ice cloud property retrievals.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:44:57 UTC from IEEE Xplore. Restrictions apply.